NATIONAL UNIVERSITY OF LESOTHO

BSc, B. Eng. Supplementary Examinations

CS3541-14 Computer Communication and Networks 1

July 2025 **Marks: 100** **Time: 3 Hours**

Instructions

1. This is a *closed-book* examination.
2. **Question 1 is compulsory.**
3. Choose ***any 4*** other questions from questions 2 to 7.
4. Each question carries 20 marks.
5. Each question **MUST** begin on a new page.
6. This question paper consists of 9 printed pages, including this cover page.

Question 1 (Compulsory)

1.1 Which of the following is a characteristic of multicast messages?
(a) They must be acknowledged
(b) They are sent to a select group of hosts
(c) They are sent to a single destination
(d) They are sent to all hosts on a network

1.2 Which dynamic addressing method for GUAs is the one where devices get their IPv6 configuration in a RA message and request DNS information from a DHCPv6 server?
(a) SLAAC only
(b) Stateful DHCPv6
(c) SLAAC with stateless DHCPv6
(d) Stateless SLAAC

1.3 Which method can be used by two computers to ensure that packets are not dropped because too much data is being sent too quickly?
(a) access method
(b) response timeout
(c) flow control
(d) encapsulation

1.4 What is the advantage of network devices using open standard protocols?
(a) competition and innovation are limited to specific types of products
(b) internet access can be controlled by a single ISP in each market
(c) network communications is confined to data transfers between devices from the same vendor
(d) a client host and a server running different operating systems can successfully exchange data.

1.5 Which interface allows remote management of a layer 2 switch?
(a) the console port interface
(b) the first ethernet port interface
(c) the switch virtual interface
(d) the AUX interface

1.6 After making configuration changes on a Cisco switch, a network administrator issues a **copy running-config startup-config** command. What is the result of issuing this command?
(a) the new configuration will be stored in flash memory
(b) the current IOS file will be replaced with the newly configured file
(c) the new configuration will be loaded if the switch is restarted
(d) the configuration changes will be removed and the original configuration will be restored

1.7 Which method allows a computer to react accordingly when it requests data from a server and the server takes too long to respond?
(a) access method
(b) response timeout
(c) flow control
(d) encapsulation

1.8 Why would a Layer 2 switch need an IP address?
(a) to enable the switch to receive frames from attached PCs
(b) to enable the switch to function as a default gateway
(c) to enable the switch to be managed remotely
(d) to enable the switch to send broadcast frames to attached PCs

1.9 Refer to the figure below. An administrator is trying to configure the switch but receives the error message that is displayed in the exhibit. What is the problem?

```
Switch1> config t
                ^
% Invalid input detected at '^' marker.
```

(a) the administrator is already in global configuration mode
(b) the administrator must connect via the console port to access global configuration mode
(c) the entire command, **configure terminal**, must be used
(d) the administrator must first enter privileged EXEC mode before issuing the command

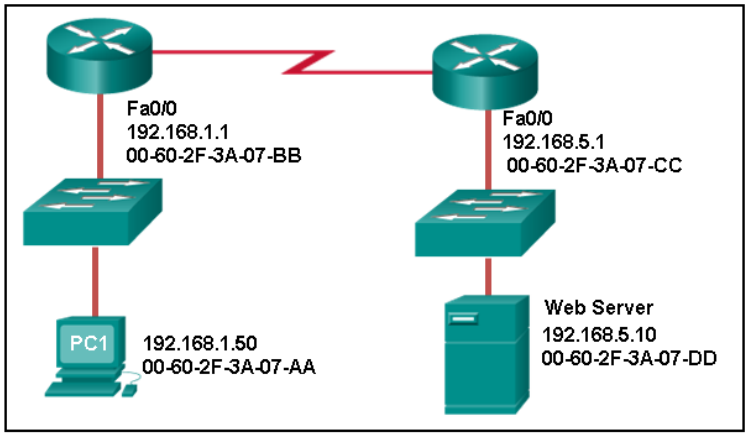
1.10 Which of the following is a characteristic of the LLC sublayer?
(a) it provides delimitation of data according to the physical signaling requirements of the medium
(b) it defines software processes that provide services to the physical layer
(c) it places information in the frame allowing multiple Layer 3 protocols to use the same network interface and media
(d) it provides the logical addressing required that identifies the device

1.11 Which statement is true about CDP on a Cisco device?
(a) because it runs at the data link layer, the CDP protocol can only be implemented in switches.
(b) to disable CDP globally, the **no cdp enable** command in interface configuration mode must be used.
(c) CDP can be disabled globally or on a specific interface.
(d) the show CDP neighbor detail command will reveal the IP address of a neighbor only if there is Layer 3 connectivity.

1.12 What type of communication rule would best describe CSMA/CD?
(a) message encoding
(b) message encapsulation
(c) access method
(d) flow control

1.13 What is the purpose of the FCS field in a frame?
(a) to determine if errors occurred in the transmission and reception
(b) to compute the CRC header for the data field
(c) to obtain the MAC address of the sending node
(d) to verify the logical address of the sending node

1.14 Carefully study the figure below.



What is the destination MAC address of the Ethernet frame as it leaves the web server if the final destination is PC1?
(a) 00-60-2F-3A-07-AA
(b) 00-60-2F-3A-07-BB
(c) 00-60-2F-3A-07-DD
(d) 00-60-2F-3A-07-CC

1.15 What process involves placing one PDU inside of another PDU?
(a) encoding
(b) segmentation
(c) flow control
(d) encapsulation

1.16 A user calls to report that a PC cannot access the internet. The network technician asks the user to issue the command **ping 127.0.0.1** in a command prompt window. The user reports that the result is four positive replies. What conclusion can be drawn based on this connectivity test?
(a) the PC can access the network. The problem exists beyond the local network.
(b) the TCP/IP implementation is functional
(c) the PC can access the Internet. However, the web browser may not work
(d) the IP address obtained from the DHCP server is correct.

1.17 Which frame forwarding method receives the entire frame and performs a CRC check to detect errors before forwarding the frame?
(a) fragment-free switching
(b) cut-through switching
(c) store-and-forward switching
(d) fast-forward switching

1.18 What is the auto-MDIX feature on a switch?

(a) the ability to turn a switch interface on or off accordingly if an active connection is detected
(b) the automatic configuration of an interface for a straight-through or a crossover Ethernet cable connection
(c) the automatic configuration of an interface for 10/100/1000 Mb/s operation
(d) the automatic configuration of full-duplex operation over a single Ethernet copper or optical cable

1.19 Why does a Layer 3 device perform the ANDing process on a destination IP address and subnet mask?

(a) to identify faulty frames
(b) to identify the host address of the destination host
(c) to identify the broadcast address of the destination network
(d) to identify the network address of the destination network

1.20 True or false? RA messages are sent to all IPv6 routers by hosts requesting addressing information.

(a) false
(b) true

[20 marks]

CHOOSE FOUR QUESTIONS FROM QUESTION 2 TO QUESTION 7

Question 2

(a) Suppose that Thabo and Liako have access to a public key system that makes their public keys available to each other. Each knows its own private key.

(i) Explain how Thabo can digitally sign a document, *m*, before he can send it to Liako. **(3)**

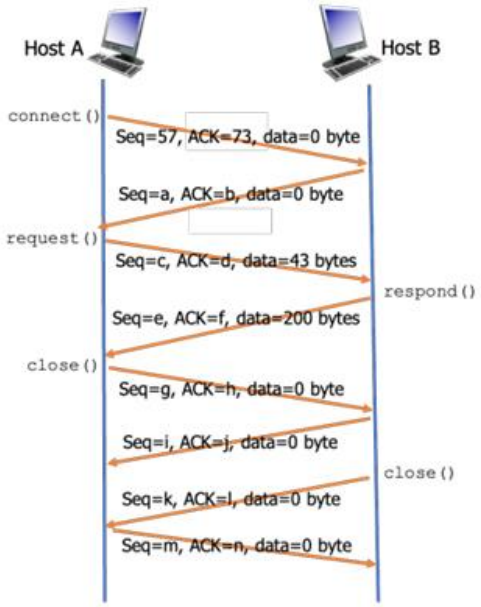
(ii) Explain what Liako needs to do to verify Thabo's signature when she receives the message. **(3)**

(b) Name the two general approaches to congestion control discussed in class.

(i) Name and explain how each works. **(4)**

(ii) Which congestion control does TCP use? **(1)**

(c) The figure below shows a sequence of packet exchange in a TCP connection between Host A and B.



(i) Briefly explain how this exchange happens. **(3)**

(ii) Calculate the values of a, b, c, d, e, g, h, j, k, l, m, n. **(6)**

[20 marks]

Question 3

(a) Explain why datagram fragmentation is not a challenge in IPv6. **(2)**

(b) Explain the role of network address translation (NAT) in internet communication. Use a diagram to show the mapping between the components involved. **(4)**

(c) Compare and contrast HTTP and SMTP along the following dimensions. Present your answer using the table given as shown below.

Dimension	**HTTP**	**SMTP**
(i) Encapsulation **(4)**		
(ii) Push vs. pull **(4)**		
(iii) Status codes **(2)**		
(iv) Message headers **(2)**		
(v) Transport protocol **(2)**		

[20 marks]

Question 4

(a) What is the role of the subnet mask when analyzing IPv4 addresses? **(3)**

(b) Given the **2001:db8:acad::/48** global routing prefix with a 16-bit subnet ID

(i) Determine the number of bits for the subnet portion of this address and write it down. **(1+2)**

(ii) Write down the first 12 subnets. **(6)**

(c) Subnet 172.16.1.0/25 to provide IP addresses for two subnets that will support the specified required number of hosts (with minimal waste). Present your answer in the table shown below as you answer all the questions (i) to (viii). **(8)**

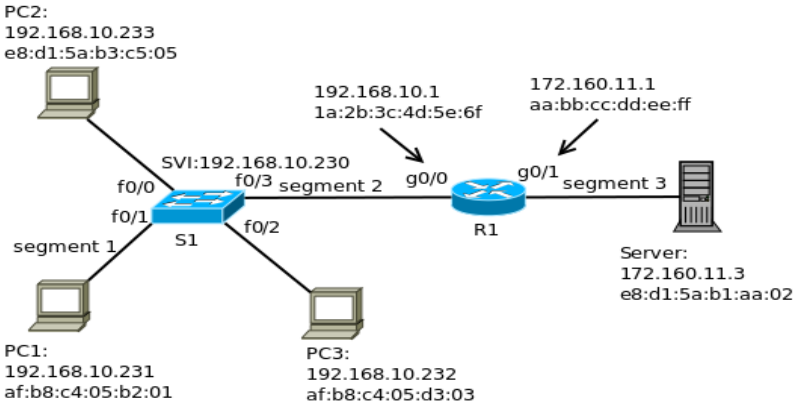
- Subnet 1: 60 hosts
- Subnet 2: 20 hosts

Specification	**Subnet 1**	**Subnet 2**
(i) Number of subnet bits		
(ii) Subnet mask (binary)		
(iii) Number of subnets		
(iv) Number of usable hosts per subnet		
(v) Subnet address		
(vi) First host IP address		
(vii) Last host IP address		
(viii) Broadcast IP address		

[20 marks]

Question 5

(a) Explain the procedure to reduce the effect of crosstalk in copper cables. **(2)**
(b) State three factors to consider when selecting a cable to implement a network. **(3)**
(c) State any three features that are present in ICMPv6 that are not present in ICMPv4. **(3)**
(d) Study the given network topology and answer the questions that follow. Assume the frames are sent from PC1 to other hosts.



(i) Indicate the values of the source and destination addresses (both MAC and IP) for segment 2 and segment 3. Present your answer in tabular format shown below. **(4)**

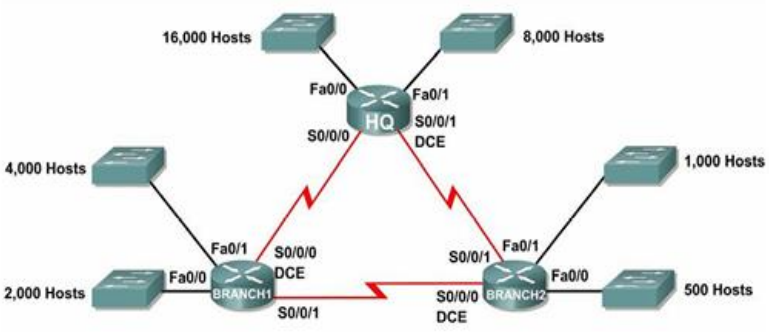
	Source		**Destination**	
Segment	**MAC**	**IP**	**MAC**	**IP**
Segment 2				
Segment 3				

(ii) Show the CAM table of S1, assuming that all hosts have communicated. **(4)**
(iii) Briefly explain the role of ARP for communication between PC1 and PC2, and between PC1 and the Server. **(4)**

[20 marks]

Question 6

(a) Apply variable-length subnetting to segment the **172.16.128.0/17** network to fulfill the IP addressing requirements of the given topology. Show your work and any assumptions you make. **(14)**



(b) Using the common IP addressing scheme, assign IPv4 addresses to all router interfaces and the switch to router interfaces. Present your answer using a table. **(6)**

[20 marks]

Question 7

(a) Define the term network protocol. **(1)**

(b) With the aid of a diagram, explain what the domain name service is and how it is implemented. **(5)**

(c) TCP is key a network protocol discussed during the course.

(i) Briefly explain the role played by TCP in a digital network. **(2)**

(ii) Name the technique used by TCP to achieve its functions. **(1)**

(iii) What type of connection does it use? **(1)**

(iv) Explain how TCP overcomes the problem of duplicate connections. **(2)**

(v) Explain how IPv4 to native IPv6 migration techniques can be implemented. You can use diagrams to illustrate. **(8)**

[20 marks]